DO-178B/C

Assurance principles for agent workflows

Reader contract	What this guide provides
Audience	Agent-workflow maintainers and engineering leads. No aviation background is assumed.
Purpose	Understand the standards' core ideas, distinguish transferable mechanisms from analogy, and interpret the local experiments.
Scope	Public-source learning and provisional review practices—not certification guidance, a compliance audit, or proof of agent safety.

Bottom line: build a defensible connection between specified behavior, relevant verification and the conclusion being claimed. Two-way review, source-first judgment and consequence-directed optional checks are implemented locally, but small exploratory comparisons did **not** demonstrate an advantage over baseline.

Evidence boundary: source research was conducted on 20 October 2026; this guide consolidates the subsequent exploratory work through 21 October 2026. Licensed normative standards were not inspected. Important authority and experiment limits remain alongside the claims they qualify.

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1 What DO-178B and DO-178C establish

DO-178B (1992) and DO-178C (2011) are editions of **Software Considerations in Airborne Systems and Equipment Certification**, paired with EUROCAE ED-12B/C. They describe software lifecycle objectives and evidence used in airborne certification, not a programming language, a mandatory waterfall method or a promise of zero defects. RTCA identifies C as the current edition and references FAA AC 20-115D as its recognition route. [1], [2]

Certification concerns a particular aircraft or system and the applicable authority's accepted means of compliance. Buying a standard, following a checklist or using a vendor tool does not itself certify a product. These public notes do not determine whether DO-178B is permitted for a particular new or modified project.

1.1 The assurance argument comes before the paperwork

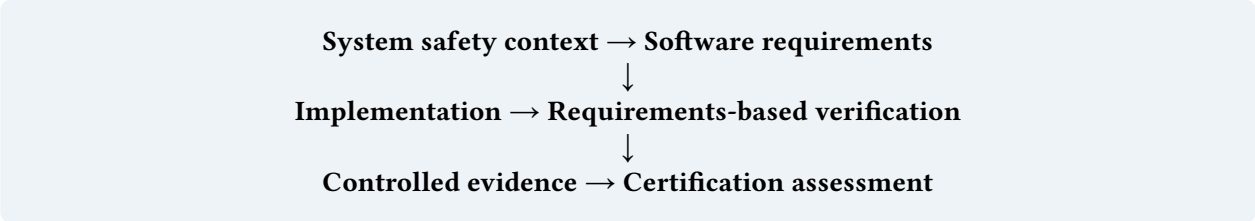


Figure 1: A conceptual assurance chain. Planning, quality assurance, configuration control and discrepancy handling support every stage; this is not a mandated development sequence.

Requirements-based verification checks intended behavior and relevant abnormal inputs against requirements, rather than merely testing whatever the implementation happens to do. **Traceability** connects requirements, implementation and tests so omissions and unjustified functionality can be investigated. **Configuration management** keeps the versions, dependencies and results belonging to a conclusion under control. [2]

These ideas were already central to B. NASA's comparison describes substantial continuity in software levels, lifecycle processes and verification; C clarifies and extends that foundation. Pothon's specialist interpretation —not regulator or normative text—also discusses clarifications to independence and reverification rather than treating them as inventions of C. [2, §II], [3, §4.3]

1.2 Rigor follows failure consequences

Software levels are assigned in the system safety context, not from code size or confidence in the developer. The following shorthand introduces the categories; it is not an assignment procedure. NASA's coverage tutorial predates C and supports B-era coverage foundations, not C-only changes. [2, §II.A], [4, §2.2]

Level	Associated failure condition	Structural coverage emphasis
A	Catastrophic	Statement, decision and modified condition/decision coverage (MC/DC)
B	Hazardous / severe-major	Statement and decision coverage
C	Major	Statement coverage
D	Minor	No structural coverage objective
E	No effect on safety	No DO-178 software assurance objectives

Statement coverage asks whether statements were exercised; **decision coverage** asks whether decision outcomes were exercised. **MC/DC** additionally asks whether each condition can independently affect its decision's outcome. These are structural measures, not a proof that the requirements are correct or that all defects have been eliminated. [4, §§2.2–2.5]

Structural coverage analysis supplements requirements-based tests. Uncovered code can indicate a missing test, a missing requirement, erroneous or extraneous code, or legitimate deactivated code; those cases require different resolution. Recording an explanation is not a license to retain unintended dead code. [3], [4]

Some verification objectives require independence from the producer of the item being checked. The separation is objective-specific: a different name, organization or model provider alone does not demonstrate that the relevant independence condition is met. [3, §4.3]

1.3 What C changed or clarified

Topic	B → C learning
Bidirectional traceability	C makes verifiable links in both directions explicit; B already had traceability intent. Forward links expose omissions; backward links expose unjustified implementation. [2, §III.C]
Additional executable code	For Level A, compiler/linker-generated code not directly traceable to source needs additional verification. C makes this objective explicit in Annex A. [3, §4.3]
Parameter data	C gives explicit attention to parameter data affecting behavior: an executable depends on more than its source files. [2, §III.D]
Tool qualification	B contained tool guidance; C uses criteria with separate DO-330 guidance. Intended use, possible errors and output checking matter more than the tool's brand. [2, §§III.E, VI], [5]
New technologies	Technology supplements add, modify or delete applicable core guidance, rather than grant blanket exemptions. [1]

The principal trace relationships include system-to-software requirements, high-to-low-level requirements, requirements-to-code and requirements-to-test cases, followed by cases-to-procedures and procedures-to-results. "B had no traceability" is therefore an incorrect account of the change. [2, §III.C]

1.4 Companion documents and their boundaries

Document	Role
DO-248C	Supporting information, FAQs and discussion papers
DO-330	Standalone software tool qualification guidance
DO-331	Model-based development and verification supplement
DO-332	Object-oriented technology and related techniques supplement
DO-333	Formal methods supplement

Formal methods can satisfy specified verification activities when their assumptions and remaining obligations are addressed. They are neither a universal replacement for testing nor categorically barred from replacing particular activities. A tool qualification argument is likewise bounded by the functions, version, operational environment and certification use involved; a qualification kit is not blanket qualification of every feature. [2, §IX], [5]

2 Transfer the principles, not aviation bureaucracy

The useful local question is whether a review connects requested behavior, actual changes and sufficient observations to a justified conclusion. Aircraft failure classifications, certification roles and coverage obligations do not transfer wholesale to nondeterministic agents. The mechanisms below are local research proposals, not findings established by the aviation sources.

2.1 Start from what the setup already does

Before this work, the setup already required criteria derived from requirements, separate reviewers for applicable work, revision-bounded evidence and explicit unresolved outcomes. It also required checking an action's intended result rather than treating tool success as completion. The opportunity was to improve how those controls are applied, not to invent an absent assurance system.

Assurance principle	Agent-workflow interpretation
Requirements first	Derive checks from the user's cumulative request and applicable safety constraints, not solely from the delivered artifact.
Bidirectional tracing	Ask both what remains unsupported and what changed without justification.
Meaningful independence	Let the reviewer assess raw sources before being anchored by the author's interpretation.
Controlled evidence	Limit conclusions to the observed target and conditions; assess relevant changes before reusing a result.
Consequence-linked rigor	Direct discretionary effort toward consequential failure paths without reducing mandatory checks.

2.2 A: review in both directions

Request → result: which requested outcome remains unsupported, contradicted or untested where testing is required? This catches omissions that disappear when a review follows only the author's summary.

Change → justification: which changed behavior or side effect lacks a requirement, necessary implementation rationale or safety justification? This catches unjustified additions without assuming that every implementation detail must appear literally in the user's wording.

A private helper needed to preserve a public interface can be justified derived work. Enabling telemetry while changing a retry default is a different behavioral change and needs its own justification. Use the existing review response; no requirement ledger or unrelated legacy cleanup is needed.

Hypothesis and falsifier: these two directions should reduce accepted omissions and unjustified changes compared with current review at matched effort. No improvement—or increased rejection of justified work—would undermine that proposed benefit.

2.3 B: judge sources before the author's narrative

1. **First view:** supply the cumulative requirements, applicable constraints, artifact or action observations and all essential raw facts. Defer the author's interpretation, not facts needed to decide correctly.
2. **Provisional judgment:** record a brief source-based view in the existing response or checkpoint. A private reasoning transcript is neither required nor requested.
3. **Reconciliation:** read the author's explanation, investigate differences and revise only when the evidence warrants it.

Use the existing reviewer where possible. If the narrative was already read, is embedded in the source, or cannot be separated, disclose that limitation rather than claim blinding. A fresh context or different provider is not proof of uncorrelated error or aviation-grade independence.

Hypothesis and falsifier: changing information order may reduce anchoring on an author's plausible story. No reduction in accepted defects, or more blocking of clean work, would undermine the benefit. The local experiment exercised a fresh-worker relay, not same-session anchoring reduction; see Section 3.

2.4 C: direct optional checks toward consequences

Reserve every mandatory safety, authority, contract, independent-review and existing quality-gate check first. Within the remaining authorized budget, prefer checks likely to expose consequential failures—especially those difficult to reverse or unlikely to be observed before harm. An unclear boundary needs inspection before it can be treated as low consequence.

Example	Useful additional check
Tenant data export	A synthetic two-tenant case for cross-tenant leakage
Deletion scope	An authorized disposable dry run checking sibling paths
Refund rounding	A negative half-cent boundary, not another positive example
Help-text typo	Inspect the affected sentence, rather than unrelated money or deletion paths

These examples prioritize checks, not permission to execute harmful workloads. Use authorized disposable reproductions rather than live privacy, deletion or financial hazards. Do not assign official aviation A–E labels to agent tasks or let a risk estimate waive a mandatory gate.

Hypothesis and falsifier: consequence-directed optional checks should expose more consequential errors within the same budget. No detection benefit, wasteful escalation on clean tasks, or misclassification of important failures would challenge that claim.

2.5 Keep ideation attached to its causal question

The research-ideation skill now asks for a visible chain:

Source principle → Intervention → Mechanism
↓
User-relevant outcome → Disconfirming observation

Figure 2: A relevance check for a hypothesis, not evidence that its mechanism works.

Separate a source's causal principles from contingent examples, formats and tooling. When a leaf detail dominates, ask why it is causally necessary and whether a simpler intervention tests the same idea. After null results, revisit the hypothesis rather than adding bookkeeping or increasingly elaborate prompts.

Do not discard necessary measurement detail in the name of abstraction. A bacterial assay's dilution factor and detection limit are essential to interpreting colony counts; an archive-naming convention usually is not. First-principles stories remain conjectures until tested, and contrary evidence must survive a persuasive analogy.

Operational placement: the existing reviewer skill and its delegation guidance reference these review practices; the research-ideation entry, tournament and proposal template carry the drift check. They add no tool, ledger, mandatory reviewer or approval gate. Their implemented status does not establish their effectiveness.

3 What the exploratory experiments show

Finding: the local comparisons did not demonstrate an improvement over baseline. That is a limit on the evidence, not a reason to relabel a successful mechanism check as workflow value or tune answer keys until an intervention wins. The practices remain provisional.

3.1 Earlier work: null results and a withdrawn detour

An initial evidence-challenge smoke test completed three reviews before another job timed out: two deficient packets were blocked, while one clean packet was held for an unrequested reviewer identifier. No matched treatment comparison completed, so this observation did not establish that the challenge caused over-checking. The timeout showed caller termination, not its underlying cause.

Two subsequent development batches produced 46 packet judgments in five fresh agent jobs. All arms agreed on accept-versus-block outcomes; changing the applicability reminder showed no acceptance advantage. Differences between explicit rejection and insufficient evidence were retained rather than relabeled into a gain.

A later checksum exercise showed that standard byte checks detect edits to covered files. It also left unlisted dependencies, identical bytes under the wrong root and post-check recapture unresolved; the comparison was against a revision-label-only shortcut, not the full existing workflow. The optional recipe was withdrawn because successful byte checking had not demonstrated useful agent-level impact and had displaced the core assurance question.

3.2 Core-principle review comparison

The next experiment compared a short requirements-based baseline with each practice separately: two-way review, source-first judgment and consequence-directed optional checks. Six review fixtures covered an omitted language requirement, unjustified telemetry, a justified helper, clean read-only work, insufficient PDF-readability evidence and sufficient link-existence evidence. Four additional fixtures offered one optional check for tenant isolation, deletion scope, refund rounding or a typo.

The fixture observations were supplied facts, not live privacy, deletion or financial workloads. Expected decisions and relevant check choices were fixed before judging and independently checked. A missing required observation was scored as insufficient evidence, not as proof that the artifact itself was defective.

Observed endpoint, per arm	Baseline	A	B	C
False acceptance among six deficient or unknown cases	0	0	0	0
Unnecessary blocking among four clean cases	0	0	0	0
Relevant optional checks selected	4/4	4/4	4/4	4/4
Consequential defects exposed by selected observations	3/3	3/3	3/3	3/3

Interpretation: baseline already solved these cases, creating a ceiling. Identical counts are not proof of equivalence, safety or production effectiveness. Results within a batch are correlated, not independent replications.

All four arms used two fresh worker stages with the same optional-check budget. Only the source-first arm received the author narrative after its provisional record; the other arms received it earlier. This exercised controlled information delivery through a relay, not reduced anchoring in one continuing reviewer.

The baseline was a simplified instruction, not the complete operational workflow. Cases were conspicuous development examples, order was fixed and each arm ran once; the combined A+B+C workflow was not a fifth comparative treatment. Exact model/version, token use and billed cost were unobservable, so elapsed job time does not establish efficiency.

3.3 Research-ideation comparison

Two fresh workers applied the pre-edit or revised entry guidance to the same assurance brief, a laboratory-assay brief and an inaccessible-source control. Both avoided checksum-centered proposals, retained CFU/mL, dilution and detection-limit information, and disclosed the unavailable source. The revised response made causal links more explicit, but that is not a demonstrated gain in ideation quality.

Both responses overstated some causal mechanisms and introduced unvalidated decision thresholds. The revised response also exceeded the 1,100-word target: 1,236 whitespace words versus 1,099 before. The guard is an explicit instruction change, not a guarantee that an agent will follow it or that first-principles reasoning will succeed.

This comparison used one response per guidance version and tested the entry guidance rather than the full composed workflow. It did not validate proposed biological mechanisms or run a wet-laboratory experiment. No laboratory or aviation result should be inferred from these fictional transfer briefs.

3.4 How to use these findings

- **Retain uncertainty:** describe the practices as implemented prototypes, not proven improvements.
- **Measure real use:** after the next one or two uses, inspect missed requirements, unjustified changes, accepted defects, clean-task blocking and observable review effort in existing receipts.
- **Preserve fair comparisons:** use genuinely new tasks, fixed answer keys and matched budgets; do not reinterpret correlated cases as independent evidence.
- **Prefer less machinery:** simplify or remove practices whose overhead is not justified; a null result is not permission to manufacture a positive one.

This guide preserves the designs, aggregate observations and limitations needed to interpret the work. Superseded narratives, fixture dumps and exact execution receipts are retained in task-private research records rather than the reader-facing documentation. Those records are not needed to follow this guide; the PDF alone is not a complete reproducibility package.

4 References and source limits

Normative RTCA DO-178B/C texts were not purchased or read; the public explanations cited here do not replace the applicable licensed texts or certification authority. Direct retrieval of FAA AC 20-115D returned HTTP 403, so detailed provisions and legacy-B transition permissions remain unverified. [6]

The agent practices and experiments are local work, not NASA, RTCA or vendor findings. Provider diversity is not aviation independence; tool calls and checklist completion are not MC/DC. Neither the literature nor the local exploratory observations establish agent certification or comparative production effectiveness.

[1] RTCA, "DO-178 Software Standards Documents and Training." Accessed: Oct. 20, 2026. [Online]. Available: https://www.rtca.org/do-178/

[2] S. A. Jacklin, "Certification of Safety-Critical Software Under DO-178C and DO-278A," Conference paper ARC-E-DAA-TN5357, 2012. Accessed: Oct. 20, 2026. [Online]. Available: https://ntrs.nasa.gov/citations/20120016835

[3] F. Pothon, "DO-178C/ED-12C: Changes and Improvements." Accessed: Oct. 20, 2026. [Online]. Available: https://www.adacore.com/uploads/books/DO178C-ED12C-Changes_and_Improvements-Sep2012.pdf

[4] K. J. Hayhurst and others, "A Practical Tutorial on Modified Condition/Decision Coverage," Technical memorandum NASA/TM-2001–210876, 2001. Accessed: Oct. 20, 2026. [Online]. Available: https://shemesh.larc.nasa.gov/fm/papers/Hayhurst-2001-tm210876-MCDC.pdf

[5] Rapita Systems, "DO-330 Software Tool Qualification." Accessed: Oct. 20, 2026. [Online]. Available: https://www.rapitasystems.com/do-330

[6] Federal Aviation Administration, "Advisory Circular AC 20-115D." [Online]. Available: https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_20-115D.pdf